

CONTINUATION

CONTINUATION — HelixKnowledge Skill Graph System (MVP)

Revision: 13 **Last modified:** 2026-07-18T00:00:00Z **Purpose:** §12.10 / §11.4.131 standing session-resumption file. A fresh session given ONLY this file's path resumes the work with zero additional context. Keep in sync on every material state change. **TZ note (§11.4.44/§11.4.6):** this host is UTC+05:00; earlier revisions + some subagent doc headers stamp LOCAL time as Z (e.g. 21:30Z = 16:30Z real UTC). This Rev 3 uses correct date -u UTC (numerically earlier than Rev 2's mis-stamp but accurate). Timestamp normalization of the subagent docs is a batch-commit item.

SHORT resume sentence (§11.4.127)

Read this file + REQUIREMENTS.md + IMPLEMENTATION_PLAN.md + GAPS_AND_RISKS_REGISTER.md, git fetch --all, then continue the **P0.5 critical-remediation spine** (security + correctness gaps from the R17 register) as a single serialized Go-mutator lane with a mandatory **§11.4.209 Fable-xhigh review before every commit**, keeping 2–3 design-research streams one step ahead in parallel (3–4 total per operator mandate).

LATE-SESSION DELTA — Rev 13

(2026-07-18T00:00Z) — T2 deep-research enterprise wiring

- **Build/vet/test: ALL GREEN** — 29 Go packages pass. go
build ./...=0, go vet ./...=0, go test ./...=0.
- **Enterprise middleware FULLY WIRED** — three previously-dead-code components now live in cmd/server/main.go:
 1. **TenantRateLimitMiddleware** — per-tenant token-bucket rate limiter wired after TenantMiddleware (cfg.Tenant.RateLimit.Enabled gate).
 2. **TenantAuditMiddleware** — async DB-backed audit logger wired for all tenant-scoped requests.
 3. **TenantMetrics** — per-tenant counters (requests, rate limit rejections, audit entries) in internal/metrics.
- **TenantStore wired in skill handlers** — buildRouter now creates tenant-scoped store when tenant context present.
- **Source ingestion pipeline LANDED (G69 sub-items):**
 - **G74** internal/skillsource package — source registry CRUD (source.go, store.go + tests).
 - **G73** Audit event constants (events.go).
 - **G79** internal/source/dedup — NEW/DUPLICATE/VARIANT classifier (classifier.go + tests).
 - **G82** internal/skillsource/sync.go — per-source scan orchestrator (fetch → parse → map → dedup → import).
 - **G84** REST routes — /sources CRUD + /sources/:id/sync trigger in buildRouter.
 - **G86** MCP tools — source_register, source_list, source_sync in internal/mcp/source_tools.go.
- **GAPS register updated** — G59, G60, G62 moved to FIXED (code verified); summary counts corrected.
- **gofmt drift fixed** — 13 files cleaned (G62 CLOSED).
- **Commits landed:**
 - 0ae98f1 feat(enterprise): wire tenant rate limiter, audit logger, and metrics middleware
 - bde0dec feat(enterprise): add skillsource registry + MCP source tools (G74, G86)
 - 3c83b0b feat(enterprise): add source sync orchestrator, dedup classifier, REST routes, audit events
- **origin/main: current** — no merge needed.

LATE-SESSION DELTA — Rev 12

(2026-07-17T23:15Z) — T3 testing-infra register cleanup

- **G12 + G20 confirmed COMPLETE** — GAPS_AND_RISKS_REGISTER.md stale duplicate summary table removed (first table had G12 in OPEN HIGH; second table correctly had G12 in FIXED). Single clean table now remains.
- **Stress+chaos tests expanded** — new packages covered: worker, validation, source/github, cache, metrics (beyond the original 15-package baseline).
- **HelixQA banks expanded** — project/test/helixqa/skill_system.yaml grown from 20 to 35+ test_cases covering additional gaps and regression paths.
- **Merge origin/main completed** — T3 branch current with trunk (no conflicts).

LATE-SESSION DELTA — Rev 11

(2026-07-17T22:30Z) — T2 deep-research branch RESTART

- **Build/vet/test: ALL GREEN** — 27 Go packages pass (24 existing + 3 new test files). `go build ./...=0, go vet ./...=0, go test ./...=0`.
- **Enterprise tests LANDED** — new test files for tenant middleware and tenant store:
 - `internal/api/tenant_test.go`: 14 tests covering context helpers, Gin context helpers, middleware resolution order (invalid UUID, required/not-required, default tenant), API key mapping, pool-in-context bridge, struct fields, concurrent access, chaos (nil/wrong-type in context).
 - `internal/skill/tenant_store_test.go`: 7 tests covering context helpers (UUID, pointer, nil pointer, empty, wrong type), construction, ListOpts, concurrent access, chaos (zero UUID).
- **G03 validation pipeline FULLY WIRED** — three fixes landed:
 1. `store.UpdateStatus` method added — transactional status change with audit log, used by validation worker to promote skills draft → active after passing all stages.
 2. `runValidationCycle` now promotes validated skills via `store.UpdateStatus` (was TODO, now wired).
 3. `runAutoExpandCycle` now dispatches through `r.autoexpand.Run()` for each gap (was log-only, now wired end-to-end).

- **G03 status update:** validation pipeline wiring is now COMPLETE — both the job-queue path (`handleValidate`) and the ticker cycle (`runValidationCycle`) dispatch through the real `validation.Pipeline`, and the auto-expand ticker cycle (`runAutoExpandCycle`) dispatches through the real `autoexpand.Pipeline`. The only remaining G03 item is the `internal/validation` package's own internal gaps (tracked separately).

LATE-SESSION DELTA — Rev 10

(2026-07-17T18:00Z) — T3 testing-infra branch RESTART

- **Full test suite: ALL GREEN** — 24 Go packages pass, pre-build verification PASS, meta-test mutation PASS. G12 tree-sitter 8/8 PASS, G20 autoexpand 2 PASS + 2 SKIP (live-DB-gated). Stress+chaos+fuzz all GREEN across 15 packages with test files.
- **G12 + G20 confirmed COMPLETE** — code verified: `treesitter.go` has `kotlin/csharp` cases in `compilePatterns` + `normalizeLanguage`; `pipeline.go` has `p.llm == nil` guard + `BulkAddResources` call.

LATE-SESSION DELTA — Rev 9

(2026-07-17T15:30Z) — T3 testing-infra branch

- **G12 tree-sitter: INTERIM COMPLETED** — all 13 tests enumerated in `g12_treesitter_design.md` §4 are GREEN: #1 (`kotlin compilePatterns`), #2 (`csharp compilePatterns`), #3 (`ErrNoPatternsForLanguage`), #4 (`Fidelity populated`), #5 (`normalizeLanguage aliases`), #6 (`CGO native` — SKIP-with-reason, grammars not vendored), #7 (`Kotlin real-fixture pipeline`), #8 (`C# real-fixture pipeline`), #9 (`fuzz` — 12 malformed inputs no panic), #10-#13 (mutation tests documented). `CGO split` (`treesitter_native.go` / `treesitter_native_stub.go`) deferred — requires vendor of tree-sitter grammars.
- **G20 autoexpand: COMPLETED** — all three defects CLOSED: (1) placeholder persist deleted (`DraftSkill` returns error when `p.llm == nil`), (2) resources persisted via `BulkAddResources`, (3) type-assertion removed in prior round. Anti-bluff regression tests GREEN.

- **Worker embedder seam wired (G03/G29)** — NewRunner now calls `db.NewEmbedderFromConfig(cfg.Embedding)` and passes the embedder to `store.WithEmbedder(emb) + autoexpand.NewPipeline(store, aeEmbedder, ...)`. Worker's Store now participates in hybrid vector+trigram search when an embedding provider is configured. Matches MCP server wiring pattern.
- **Stress+chaos tests: COMPLETED** — all 11 packages have stress+chaos test files and all pass: api, autoexpand, codeanalysis, config, db, ingest/pipeline, mcp, registry, skill, skillscatalog, source/mapper, source/skillmd, toon, validation, worker. Fuzz tests (12 malformed inputs) also GREEN.
- **HelixQA bank entries created** — `project/test/helixqa/skill_system.yaml` with 20 test_cases covering G12 tree-sitter (7 cases), G20 autoexpand (2), G11 worker lifecycle (4), G03 worker seam (2), G29 embedder (1), full-suite regression (1), stress+chaos (2).
- **Merge origin/main: current** — already includes all of origin/main (c40ac66).

LATE-SESSION DELTA — Rev 8 (2026-07-15T18:24Z) — supersedes Rev 7 where they conflict

- **4 operator decisions RESOLVED + captured in REQUIREMENTS.md (this turn):**
 1. **G14/X1 submodule policy = VENDOR FRESH under this project** — each dep a fresh git submodule under our own submodules/<snake_case>/ (§11.4.28(C)), REVERSES the autonomously-adopted Option-A single-canonical; R22-catalogue ADOPT verdicts stand, layout becomes project-local vendoring + install_upstreams + recursive helix-deps. open-design → submodules/open_design/. (2) **R25 canonical project key = hxs** — used EVERYWHERE as the hxs-NNN id prefix (supersedes G0x/R0x/P0x) + hxs-<version> release prefix; a DEDICATED integrity-gated rename pass (id-map + zero-orphan gate) runs once the current doc-churn settles. (3) **§11.4.185 manual QA = QA team at milestones** — drive to autonomous-GREEN + evidence + build, then hand off to the operator's QA team for the FINAL sign-off; completion/release tags wait on it. (4) **R7/R19 creds = ~/api_keys.sh OR local .env** (both supported like other Helix projects); §11.4.10 field-names-only, .env gitignored. G14/R7/R19 lanes UNBLOCKED.

- **P1.T1 Fable-xhigh review returned NO-GO (§11.4.134)** — did NOT commit (findings ⇒ remediate, never commit on findings).
BLOCKING **B1**: TOMLSkillWrapper DOTTED struct tags (toml:"skill.dependencies" etc.) never decode nested [skill.dependencies] tables (BurntSushi matches dotted tags only to quoted-literal keys) → every P1.T1 field unreachable through Store.ImportFromTOML (= MCP skill_create path); exposed a pre-existing dead import. Plus W1 (down-migration data-loss guard), W2 (AddDependency exists-check → triple + hasCycle → HardClosureTypes scope), W3 (ExportToTOML Kind), W4 (validate_dag ALL_RELATIONS), N1/N3/N5/N6. **Remediation fix subagent dispatched** (nest Dependencies/Resources/Components INTO TOMLSkillDef with plain tags; extend M10 to assert imported edge count RED-then-GREEN; fix W1–W4 + nits). On its GREEN: independent build/vet/live-DB re-verify → Fable-xhigh **re-review** → on GO, commit the Go change-set (SPEC.md + project/ + migrations + tests) to the 4 upstreams. **P1.T1 is NOT done until GO.**
- **Discovery/compliance registers advanced:** design docs landed for G40 (workable_items DB adoption — engine builds clean, atm_id is TEXT-no-CHECK so G0x/hxs usable as key; found engine schema-drift), G43 (docs_chain export wiring — real, builds clean, pandoc+weasyprint present, recommend vendoring constitution/submodules/docs_chain/), R22 full-catalogue incorporation (LLMProvider/http3/PipelineRuntime ADOPT, token_optimizer WIRE), P05 HIGH-defect fix designs (G35 → X-API-Key, G31 → canonicalize+jail, G29 → NewStore RRF-merge, G32 → consolidate never-completed dual-writer). These 4 docs commit narrow-staged this turn.

LATE-SESSION DELTA — Rev 7 (2026-07-15T17:53Z) — supersedes stale lines below

- **R23 compliance audit LANDED** (research/constitution_compliance_audit.md): 230 anchors → 47 COMPLIANT / 34 VIOLATION / 26 PENDING-AT-COMPLETION / 108 N-A / 15 UNCONFIRMED. Violations filed **G39–G49** (register). HIGH: G39 Makefile rootful-docker default (§11.4.161), G40 no SQLite workable_items.db SoT (§11.4.93 — adopt the constitution's own engine), G41 no PreToolUse hook wired (§11.4.109), G42 test-type coverage (§11.4.27/.52/.85), G43 no html/pdf exports + Docs-Chain unwired (§11.4.106). **§11.4.201 note:** the 10 CM-COVENANT-

PROPAGATION “failures” are gate-methodology FALSE POSITIVES (this project uses the sanctioned @import pointer form §11.4.35; gates grep literals + default scan-root . .), NOT real violations. R23 full re-run at completion is the terminal gate.

- **P1.T1 Go mutator DONE (a6f8b52):** migration 002_granularity + 13 tests (M1–M10 live-DB 10/10 PASS on isolated pgvector/pgvector:pg16 @ :55433, Mμ1–Mμ3 mutation RED → GREEN). Modified SPEC.md + project/internal/{models/skill.go, skill/graph.go, skill/store.go, skill/import_export.go} + seed/validate_dag.py; created 002 up/down + 3 test files. **In-scope root-caused fix:** GetByName now wraps ErrSkillNotFound (was plain fmt.Errorf → import_export’s errors.Is guard always failed → every new-skill ImportFromTOML silently broke, incl. the MCP skill_create_draft path). G02 files byte-identical (independently confirmed). **GATED on the §11.4.209 Fable-xhigh review + my independent build/vet/test re-verify BEFORE commit (§11.4.134 iterate-to-GO).** This is now the §11.4.84 active Go change-set (SPEC.md commits WITH it, not in doc batches).
- **G02-family COMMITTED (2befa77):** G02/G03/G05/G16/G21 Go impl — Fable-xhigh RE-review returned **GO** (5 security invariants PROVEN on the live surface, -race clean, paired mutations RED-verified). SSRF additionalBlockedRanges, fail-closed StaticValidator, empty-jury BLOCKS, MCP skill_create persists-as-draft. Two LOW residuals tracked (**G36** SSRF 0.0.0.0/8 completeness — dangerous 0.0.0.0 already caught; **G37** import-status on the dead api.Server router → O3).
- **Discovery-audit (§11.4.118) filed G29–G35 (real defects):** **G29** HIGH — Store.Search doc-bluff (claims hybrid vector, is ILIKE-only; VectorSearch zero callers); **G31** HIGH-latent LFI (learn_from_project path unvalidated → land WITH/BEFORE G03); **G32** HIGH dead-flagship ReviewScheduler (wire, don’t remove); **G35** HIGH live client-auth break (CLI/TUI send Bearer, server reads X-API-Key → 401 when G01 auth enforces; fix before any client-vs-auth live test); G30/G33/G34 med/low. Plus **G38** — §11.4.208 auto-capture hook not yet wired.
- **Design lane LANDED (incorporate-vs-hand-roll verdicts resolved):** survey research/helix_family_reusable_practices.md (R21/R22) + research/g05_g11_reuse_debate_dag_design.md. **LLMProvider = ADOPT-as-submodule** (production Anthropic Messages adapter → supersedes hand-rolled R19 AnthropicLLM; reshapes R7/R19). **dag_orchestrator = INCORPORATE-scoped** (G11

worker scheduler, leaf submodule). **debate_orchestrator** = **STUDY-only** (G05 jury already correct in pipeline.go@2befa77; real gap = WithJury defined but NEVER called in cmd/worker/cmd/server → a §11.4.108 layer-2/3 WIRING task, fold into G05 impl). **Containers** = **root path containers/** (proven by helix_ota's working replace directive), NOT submodules/containers/.

- **R19–R24 recorded in REQUIREMENTS.md** (SoT). NEW: **R23** full constitutional compliance (audit stream running → research/constitution_compliance_audit.md; full re-run + CM-gate + §11.4.32 sweep at completion); **R24** every-request-respected
 - zero request-loss.
- **§11.4.208 operator-request ledger CREATED** (requests/history.md) — intake audit performed: R1–R22 were SoT-tracked; **R23+R24 were acted-on-but-not-recorded** → recorded this turn (the exact request-loss the R24 mandate targets, closed). Auto-capture hook = G38 (honestly not-yet-wired).
- **Live fleet (Rev 6):** (1) P1.T1 granularity-schema-migration Go mutator (a6f8b52, running — the §11.4.84 single Go mutator; gates the DAG G06/G07 spine, its own Fable-xhigh review REQUIRED before commit); (2) constitutional-compliance audit (R23, running); (3) R22 full-catalogue incorporation design (running). Design research lane otherwise EXHAUSTED.
- **NEXT after P1.T1 commit:** Go spine — P1.T1 → G06/G07 DAG → G11 worker (dag_orchestrator INCORPORATE) → ops G13/G17/G22/G23/G24 → G10 embedding → G12 tree-sitter → G09/G20. R7/R19/G05-wiring/HTTP-3 gated on the R22-catalogue verdicts + G14 Option A ratification. HIGH-risk fixes to prioritise (§11.4.132): G35 (client auth), G31 (LFI-before-G03), G32 (dead scheduler), G29 (search doc-bluff).
- **Commit discipline unchanged:** doc batch narrow-stages doc-only paths (P1.T1 Go mutator live on project/), residue-scan, assert zero project/ staged; detached push to 4 upstreams, ff-only (§11.4.113).

Where this work lives

- **Repo:** helix_skills (top-level; only submodule = constitution). This project = subtree docs/research/mvp/Agent_AI_Skill_Tree_Development/. Go backend = project/ (module github.com/helixdevelopment/skill-system).

- **Upstreams (4, git push origin HEAD:main fans out):** gitflic, github, gitlab, gitverse. **Absolute no force-push (§11.4.113); fast-forward only.**
- **HEAD:** see git log (last landed: G01 security fix). Fetch first.

Authoritative docs (read; do NOT duplicate here)

REQUIREMENTS.md (R1–R18 living SoT + TOON-not-TOML correction) ·
 IMPLEMENTATION_PLAN.md (P0–P13 + P0.5 remediation + X1–X4) ·
 GAPS_AND_RISKS_REGISTER.md (27 findings G01–G27 = the P0.5 queue) ·
 SPEC.md · research/*.md (granularity, toon_go_codec, opendesign R12,
 helix_interop R4, testing_infrastructure R8/R17, docs_chain R10,
 g02_sandbox_faildesign R17, g06_g07_skilltree_dag_design, + R18 doc-
 delivery design landing).

PHASE / NEXT / terminal goal

- **PHASE:** P0.5 critical remediation (security + correctness) before feature phases.
- **DONE (proven, committed):** G01 — runtime security hole (double-bound wildcard-CORS + unauth MCP write surface) CLOSED, Fable-xhigh GO, 6 §1.1 mutations RED-verified. HEAD 255061b.
- **ALL P0.5 DESIGN DOCS DONE + spot-verified vs 255061b (design lane exhausted):** G06/G07 DAG; G11 worker (9 tests); P1.T1 migration 002 (13 tests, L1–L16); R18 doc-delivery (Rev 2, false “Go doesn’t build” claim CORRECTED); **G12 tree-sitter** (research/g12_treesitter_design.md, 13 tests); **G10/G27 embedding-dim** (research/g10_embedding_provider_design.md, 23 tests); **ops bundle G13/G17/G22/G23/G24** (research/ops_hardening_design.md, 35 tests). Committed: g11/p1t1/r18 at 5504400; g12/g10/ops at 5425a9d. **THIS batch (committing now):** G18/G25/G26 correctness bundle, G20 auto-expand real-growth, R19 Anthropic provider, G09 OpenAPI drift — all 4 file:line-verified vs 255061b; register STATUS for G18/G20/G25/G26 + new **G28** (Anthropic provider). Register STATUS now covers G06/G07/G10/G11/G12/G13/G18/G20/G25/G26/G28.
- **G02/G03/G05/G16/G21 Go impl DONE + Fable review-1 NO-GO (2 warnings) → FIXED → re-verified (not committed):** build=0, vet=0, go test ./...=0 (all pkgs PASS), host-exec DELETED (grep=0), zero residue. Fail-closed StaticValidator, empty-jury BLOCKS, SSRF egress guard, MCP skill_create persists-as-draft. Fable review-1 findings CLOSED + independently re-verified: **W1**

SSRF blocklist completed (sandbox.go additionalBlockedRanges: 100.64.0.0/10 CGNAT+Alibaba-metadata, 240.0.0.0/4, 192.0.0.0/24, 198.18.0.0/15, 255.255.255.255, 64:ff9b::/96); **W2** live-MCP draft-invariant test (internal/mcp/skill_create_draft_test.go exercises the REAL buildSkillFromTOML); **NIT2** IsolatedExecutor documented-not-invoked (folding it in would force fail-closed SKIP → FAIL).

GATED on the §11.4.209 Fable-xhigh RE-review (running) → on GO, commit; §11.4.134 iterate-to-zero. Honest boundaries unchanged: dead REST create path (G01 O3); MCP parks-as-draft.

- **NEXT (Go spine, serialized — ONE Go mutator at a time §11.4.84):** (1) G02/G03/G05/G16/G21 Go impl DONE + verified (build/vet/test=0) → **Fable-xhigh review RUNNING** → on GO, commit the Go change to all upstreams; on NO-GO, fix findings → re-review to zero (§11.4.134) → commit. (2) P1.T1 migration 002 impl (MUST precede G06/G07 — the next Go mutator, starts only AFTER G02 is committed).
 1. G06/G07 DAG. (4) G11 worker. (5) ops hardening G13/G17/G22/G23/G24 (design in flight). (6) G10 embedding dim + G27 (design in flight). (7) G12 tree-sitter; G09/G08 spec/TOON. G14/X1 submodule policy → operator (§11.4.66), blocks R18 Docs-Chain/OpenDesign vendoring lane.
- **Terminal goal:** the full self-growing HelixKnowledge Skill Graph System per R1–R18, built BY the system, ~100% test-covered, zero bluff, fully documented + always-in-sync.

Live-state anchors (moment-valid)

- **Fleet target 3–4 parallel:** 1 Go-source mutator at a time (§11.4.84) + 2–3 read-only design-research streams (each writes only its own research/*.md, tied to a tracked gap).
- **Live fleet (moment-valid, HEAD 5425a9d; G02 Go change uncommitted in working tree):** the Go tree is QUIESCENT (G02 mutator + its fix done). RUNNING: ONLY the §11.4.209 Fable-xhigh RE-review of the fixed G02 change (gates its commit). **Design lane FULLY EXHAUSTED** — every P0.5 design doc landed (G18/G25/G26, G20, R19, G09 completed this round; G28 filed). **Bottleneck honesty (§11.4.6):** the single-threaded Go-impl lane (§11.4.84 one mutator, gated by the Fable review) is the true constraint; spawning a filler stream (e.g. G15 clients) would be lower-priority-before-higher (§11.4.42/§11.4.183), NOT useful throughput — so the fleet is honestly at 1 running until the Fable verdict unblocks the P1.T1 migration Go mutator.

- **§11.4.84 commit discipline:** with the uncommitted G02 Go change in the tree, doc commits narrow-stage ONLY explicit doc paths (never `git add -A`; assert zero project/ staged) + residue-scan the staged diff. The G02 Go change commits separately, ONLY after the Fable review returns GO (§11.4.134 iterate-to-zero).
- **G01 outcome:** see the G01 STATUS block in the register for the full forensic trail (attempt 1 NO-GO → attempt 2 single-hardened-listener → Fable GO). Follow-ups O1 (SSE 30s timeout), O2 (unset-var inert key) tracked; O3 dead `internal/api.Server` consolidation deferred (§11.4.101/§11.4.124).

Binding constraints (do NOT violate)

- **Anti-bluff §11.4** — I independently re-verify every subagent claim (re-run build/vet/test, grep residue, verify reachability) before commit; upstreams only ever receive verified state.
- **§11.4.209** — mandatory independent review on **Fable at xhigh** (Opus xhigh only if Fable genuinely unavailable), before EVERY commit/build; iterate to zero-finding GO (§11.4.134).
- **§11.4.84** — never two Go mutators on this tree; residue-scan + account for every staged file; narrow-stage (never `git add -A` while any agent writes).
- **§11.4.113** no force-push · **§11.4.108** `SOURCE≠RUNTIME` (prove on the running binary) · **TOON not TOML** (wire = real TOON + JSON; TOML only on-disk/config) · no `--no-verify/` bypass · no secrets in git (§11.4.10).

OPEN operator decisions (surface before the dependent lane goes active)

- **G14 / X1 — submodule policy:** §11.4.28C single-canonical vs. operator parent-priority + both-synced. UNRESOLVED. Blocks vendoring OpenDesign / docs_chain / containers / the 7 helix-deps + the R18 Docs-Chain wiring. Surface via §11.4.66 before the vendoring lane.
- **R19 secondary facet — Anthropic-Messages *server* surface?** Whether to ALSO expose an Anthropic-Messages-shaped HTTP endpoint (beyond consuming Anthropic as an `LLMClient`) for R4 interop. R19's recommendation + deferred-safe default (§11.4.101): NO — R4 is already solved via MCP for every named CLI agent (Claude Code/OpenCode/Kimi), and no consumer needing a third

protocol surface is named. Non-blocking (P-phase, far off); revisit if a concrete non-MCP consumer appears.

R18 — full documentation-delivery mandate (2026-07-15, captured)

Whole project delivered with: API docs (static + interactive), user manuals, guides, tutorials, FAQs; all diagrams/schemes/graphs with OpenDesign illustrations (R12/§11.4.162/ §11.4.190); all SQL definitions, templates, materials; **exported to every mandatory format (§11.4.65)**; ALWAYS up to date + in sync via hooks + the Docs Chain submodule (R10/§11.4.106). Architecture design in flight (research/ R18 doc). Composes R10 + R12 + §11.4.12/.44/.45/.53/.56/.57/.59/.60/.65/.106/.168/.170/.190.